

Deval Thakkar

Senior SDET

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SUMMARY:

Senior SDET with 5+ years of experience architecting scalable automation platforms, backend APIs, and distributed systems using Python, FastAPI, AWS, and LangChain for LLM-driven workflows testing with (Langtest,Ragas,Pytorch) Proven expertise in system design, CI/CD optimization, and async processing for high-volume data (500K+ records). Delivered robust, production-grade solutions leveraging Gen AI (RAG) for dynamic validations and real-time insights—now targeting engineering roles focused on high-impact, AI-powered product delivery.

Skills

Languages:Python (Advanced), SQL (Intermediate), Java (Basic), JavaScript (Basic)

Backend & Testing:FastAPI, Pytest, Selenium, REST APIs, Postman, Swagger, Gen-AI[LangChain, RAG pipelines, LLM APIs (OpenAI, Ollama)] LangTest,Ragas

Cloud & DevOps: AWS (S3, Lambda, EventBridge, SageMaker, CodePipeline), Docker, Jenkins, kubernetes,Vector Databases (Chromadb), CI/CD tools, Git

works & Tools:Async Processing, Microservices, Redis, PySpark (Basic), PyTorch, Prompt Engineering, Prompt Chaining, Hugging Face Transformers (basic), Fine-tuning LLMs (basic), TestCollab, Test Case Management Tools (Xray–Jira)

Core Competencies: System Design, Distributed Systems, CI/CD, Big Data Validation, Test Strategy, LLM-driven Automation, Retrieval-Augmented Generation (RAG)

Other: Shift-left Testing, , CAPA/GDPR test compliance,Excellent Interpersonal Communication

Work Experience

2024-04 - 2025-03

Sr Software Engineer

Zucitech Software Solution Pvt Ltd , Bengaluru

- Architected a Pytest + FastAPI framework with advanced trackbacks handling, cutting runtime by 50% and debugging time by 60%.
- Designed and deployed an async Redis-backed pipeline processing 500K+ records, reducing batch time from 7 days to 1 hour (6x throughput gain).
- Automated CI/CD using Jenkins, Docker, and Linux environments, eliminating 70% manual steps and accelerating deployments.
- Worked with PyTorch-based ML/LLM models for inference and basic validation, ensuring integration into production APIs and data workflows.
- Delivered RESTful reporting APIs and dashboards, reducing defect triage time by 70% and enabling real-time insights.
- Conducted end-to-end LLM testing using LangTest (bias, consistency, hallucination checks) and RAGAS (retriever relevance, retrieval+generation synergy testing), ensuring robust and production-grade LLM workflows.

2023-01 - 2023-10

Lead Software Developer(QA)

Altencalsoft Lab (India) P Ltd, Bengaluru

- Built a FastAPI test automation framework integrated with Xray–Jira, boosting coverage by 30% and cutting manual QA by 50%.
- Automated deployments via linux based AWS CodePipeline, CodeBuild, and Docker, improving release stability and speed by 40%.
- Integrated LangGraph and RAG pipelines to validate complex data workflows and enable dynamic API testing.

- Conducted end-to-end LLM testing using LangTest (bias, consistency checks) and RAGAS (retriever and generation accuracy).
- Engineered a layered Selenium + Pytest system, increasing defect detection by 25% in test flows.
- Conducted testing of PyTorch-based ML and LLM models,(Inference correctness,Integration), ensuring they met expected output and integration with other systems.
- Implemented Grafana-based 24/7 test observability and ensured CAPA/GDPR compliance.

2021-06 - 2023-01

Senior QA Engineer

Nolan Edutech Pvt Ltd, Bengaluru

- Built a Selenium-backed API and DB validation suite for Masai onboarding, cutting manual verification by 70% and improving data accuracy.
- Automated end-to-end onboarding with Python + FastAPI, reducing process time from 3 days to 1 day and boosting verification success by 80%.
- Integrated AWS CodePipeline with SQL-based data validation for complete backend-to-database test automation.

2020-11 - 2021-05

QA Specialist

CodeJudge It Solution India Pvt Ltd, Bengaluru

- Built automation suites using Python + Pytest, improving functional test efficiency by 50% across APIs and workflows.
- Designed a scalable API test framework with Locust for load testing and Selenium (Java, POM) for UI flows, reducing manual effort by 60%.
- Integrated test pipelines via AWS CodePipeline + Lambda, automating regression and reducing release cycles by 30%.

Projects & Accomplishment

- **Scalable API Development(Fastapi) Automation (pytest,async) (Disney OTT):**Architected a Redis + Kafka-backed async API batching system with worker queues, processing 500K+ OTT records per job; reduced latency by 90%, integrated Locust for load profiling, and scaled with AWS Lambda using event-driven microservices principles.
- **Parental Control Validation (Disney OTT):**Built a high-throughput data compliance system validating millions of access control records, reducing backend load by 40% through optimized schema parsing and parallel pipelines; aligned with GDPR/CAPA and integrated alerting for policy breaches.
- **GenAI Compliance Insights Engine (OTT, Disney OTT Data):** Architected a LangChain-powered RAG pipeline to generate compliance insights and policy summaries from OTT logs. Deployed an async FastAPI microservice using LangGraph for scalable LLM-based retrieval (ChromaDB) and integrated Slack alerts for policy breaches, reducing manual oversight by 50%.
- **No-Code/Low-Code Framework (Eli Lilly):** Engineered a domain-agnostic, drag-and-drop automation platform using Selenium POM (Java) and YAML-driven test specs, cutting QA cycle time by 60% and empowering non-technical teams to build, deploy, and maintain end-to-end test flows without code.
- **Onboarding Pipeline (Masai School):** Built a robust onboarding pipeline integrating UI validations (Selenium), backend checks, and CI/CD via AWS CodePipeline, reducing manual setup by 40% and boosting data verification accuracy to 99%+ across real-time student data ingestion.
- **Fleet Management CRM (Microsoft CRM):** Automated backend and UI workflows for 50+ fleet operations in a custom Microsoft CRM system, enabling seamless regression via Dockerized test execution and API/UI test fusion; delivered end-to-end test reliability for dynamic vehicle and inventory modules.

Award & Certifications

2023-01

Google Certification of Python

2024-07Employee of the Month

2025-02Pyspark in Action (coursera)

Education

2021-12 - 2024-06

B.E: Computer Science

Dr. A.P.J. Abdul Kalam University University - Indore

CGPA:- 8

Relevant Coursework: Algorithms, Distributed Systems, Cloud Computing

2018-06 - 2021-06

Diploma: Computer Science

Dr. A.P.J. Abdul Kalam University University - Indore

CGPA:-7